BMC Ophthalmology
https://doi.org/10.1186/s12886-026-04891-w

Article in Press

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Received: 9 February 2026

Accepted: 4 May 2026

Published online: 11 May 2026

**Cite this article as:** Li H., Jiang C., Hu X. *et al.* Dose-response association between metabolic abnormality burden and intraocular pressure in a population undergoing routine health examinations: a cross-sectional study. *BMC Ophthalmol* (2026). https://doi.org/10.1186/s12886-026-04891-w

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# Dose-Response Association Between Metabolic Abnormality Burden and Intraocular Pressure in a population undergoing routine health examinations: A Cross-Sectional Study

Short title:Metabolic Abnormalities and Intraocular Pressure

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## Abstract

**Objective:** To investigate the association between metabolic risk burden and intraocular pressure in individuals undergoing routine health examinations, and to assess the presence of a dose-response relationship, thereby informing IOP risk stratification in health check-up settings.

**Methods:** This cross-sectional study included 2,048 adults who received health examinations at the Health Examination Center of the Army Medical Center between September 2024 and December 2025. Metabolic abnormalities were defined using five routinely collected components-waist circumference, blood pressure, fasting plasma glucose, triglycerides, and high-density lipoprotein cholesterol-and were summed to generate a metabolic risk burden score (range:0-5). Participants were categorized into four groups (0, 1, 2, and⩾3). The primary outcome was the mean IOP of both eyes. Associations were evaluated using one-way analysis of variance, multivariable linear regression, and tests for trend.

**Results:** IOP increased progressively with higher metabolic risk burden. After adjustment for age and sex, compared with participants with a score of 0, mean IOP was higher by 0.93, 1.24, and 1.80 mmHg in those with scores of 1, 2, and⩾3, respectively (all *P*<0.001). Further adjustment for body mass index attenuated the associations, but they remained statistically significant (β=0.77, 0.93, and 1.37 mmHg, respectively).Each one-category increase in metabolic burden was associated with a 0.46-mmHg higher IOP (P-trend<0.001). In sensitivity analyses, higher metabolic burden was associated with increased odds of high IOP (⩾3 vs 0: OR=3.05, 95% CI: 1.73-5.36). Among individual components, elevated blood pressure and elevated fasting glucose showed the strongest associations with IOP.

**Conclusions:** In this cross-sectional study, metabolic risk burden was independently associated with higher IOP in a dose-response manner. These findings suggest a graded relationship between metabolic abnormalities and IOP; however, causal inference and clinical implications require confirmation in longitudinal studies.

**Keywords:** intraocular pressure; metabolic risk burden; metabolic syndrome components; health examination; cross-sectional study

## Introduction

Glaucoma comprises a group of disorders characterized by progressive optic neuropathy and corresponding visual field loss, and its end stage can result in irreversible blindness, posing a substantial global public health challenge[1].Epidemiological projections estimate that the number of individuals affected by glaucoma will reach 111.8 million worldwide by 2040[2]. Intraocular pressure (IOP) refers to the pressure exerted by the intraocular contents-including the lens, vitreous body, uveal tissues, retina, and intraocular fluids-against the ocular wall. It is a key clinical parameter for assessing glaucoma risk and monitoring disease progression, and remains one of the most important modifiable risk factors for glaucoma[3,4]. In health examination settings, IOP screening is routinely incorporated into basic ophthalmic assessments. With the widespread adoption of non-contact tonometry in routine check-ups, leveraging health examination data to identify individuals at risk of elevated IOP and to guide further evaluation and follow-up has become increasingly relevant for earlier detection of glaucoma[5].

Beyond ocular determinants such as refractive status and central corneal thickness, increasing attention has been directed toward the relationship between systemic metabolic disturbances and IOP. Several epidemiological studies have suggested that obesity, dyslipidaemia, elevated blood pressure, and impaired glucose metabolism are each associated with higher IOP[5-8]. Metabolic syndrome (MetS) is a clinical constellation characterized by the clustering of metabolic abnormalities, including dyslipidaemia, central obesity, insulin resistance, and hypertension. It affects more than 1.5 billion individuals worldwide, and accumulating evidence indicates that MetS is associated with elevated IOP and an increased risk of glaucoma[1,5,9,10]. Nevertheless, heterogeneity in MetS definitions, measurement approaches, and study populations limits comparability across studies[11,12].In addition, there remains no consensus on a practical and widely applicable risk stratification strategy for health check-up settings.In several studies, MetS has been treated as a binary exposure ( "present" vs "absent" ), which facilitates a broad comparison but may obscure the potential graded relationship as metabolic abnormalities accumulate from mild to severe[11-13]. Although prior research has reported a close association between metabolic risk burden and higher IOP, real-world evidence derived from routine health examination data is still relatively limited[1,5].To more directly capture the extent of metabolic clustering in relation to IOP, we therefore applied a "metabolic risk burden" (i.e., a metabolic burden score) to stratify individuals undergoing health examinations.

In this study, we constructed a metabolic burden score using routinely collected health check-up components-waist circumference, blood pressure, triglycerides, high-density lipoprotein cholesterol and glucose, grouped participants according to metabolic risk burden (0-5 points). We then evaluated the association between metabolic risk burden and mean IOP of both eyes, and to assess the presence of a dose-response relationship, thereby informing IOP risk stratification in health check-up settings.

## Study Design and Data Source

This cross-sectional study was conducted using de-identified health examination data from the Health Examination Center of the Army Medical Center. The study period spanned from September 2024 to December 2025. The study protocol was approved by the Ethics Committee of the Army Medical Center(Approval No.2026-043) , and all procedures were carried out in accordance with the Declaration of Helsinki.The dataset included demographic characteristics (age and sex), anthropometric measurements (waist circumference and body mass index), blood pressure, laboratory variables (including fasting plasma glucose, triglycerides, and high-density lipoprotein cholesterol), and intraocular pressure measurements obtained by non-contact tonometry. To minimize correlation bias arising from repeated observations, if an individual underwent more than one health examination during the study period, only the first record was retained for analysis.

### Inclusion and Exclusion Criteria

Inclusion criteria were as follows:(1)age≥18 years;(2)completion of non-contact IOP measurements in both eyes at the index examination, with IOP values available for both eyes;(3)availability of all variables required to construct the metabolic risk burden score, including waist circumference, systolic and diastolic blood pressure, triglycerides (TG), high-density lipoprotein cholesterol (HDL-C), and fasting plasma glucose.Exclusion criteria included:(1) a prior diagnosis of glaucoma;(2) current use of corticosteroids (topical ocular or systemic);(3) IOP recorded in only one eye;(4)missing data for key variables; and clearly implausible IOP values, defined as a mean binocular IOP <5 mmHg or >30 mmHg.

### Measurements

IOP was measured as part of the routine health examination using a non-contact tonometer (TX-20P; Canon Inc., Tokyo, Japan) by trained medical staff. Each eye was measured three times, and the mean value was recorded. For participants with measurements in both eyes, the mean of the two eyes was used as the primary outcome. Participants with IOP recorded in only one eye were excluded.

### Construction and Categorization of Metabolic Risk Burden

We constructed a metabolic risk burden score based on components related to MetS. Metabolic risk burden was prespecified as the primary exposure and was defined according to commonly used Chinese criteria for MetS[14,15].The five metabolic abnormality components were defined as follows:

1. Central obesity: waist circumference≥90 cm in men or≥85 cm in women;
2. TG: TG≥1.7 mmol/L;
3. HDL-C: HDL-C <1.04 mmol/L;
4. Elevated blood pressure: systolic blood pressure≥130 mmHg and/or diastolic blood pressure ≥85 mmHg, or a prior diagnosis of hypertension with current treatment;
5. Elevated fasting glucose: fasting plasma glucose≥6.1 mmol/L (participants with random glucose measurements were excluded), or a prior diagnosis of diabetes with current treatment.

The metabolic risk burden score was calculated as the sum of these five components, yielding a total score ranging from 0 to 5.

## Statistical Analysis

All statistical analyses and figure generation were performed using IBM SPSS Statistics (version 29.0.2.0) and GraphPad Prism (version 10.4.0) on de-identified data. Normality was assessed for continuous variables. Variables with an approximately normal distribution are presented as mean± standard deviation,whereas non-normally distributed variables are summarized as median (interquartile range).Estimated marginal means of IOP (EMM± standard error) across metabolic risk burden categories (scores 0, 1, 2, and⩾3) were derived using a general linear model, with pairwise comparisons conducted using Bonferroni adjustment. Multivariable linear regression models were then fitted to quantify the association between metabolic risk burden categories and IOP, and a test for trend was performed by modeling the burden category as an ordinal variable. For sensitivity analyses, high IOP was defined as an IOP value at or above the 90th percentile (P90) of the study population, and logistic regression was used to estimate odds ratios.In addition,the five metabolic components were entered simultaneously into a multivariable model to evaluate their independent contributions, and variance inflation factors (VIFs) were reported to assess multicollinearity. A two-sided P value <0.05 was considered statistically significant.

## Result

A total of 2,048 participants were included in the final analysis and were categorized into four groups according to the metabolic abnormality score: score 0 (n=1,003, 49.0%), score 1 (n=527, 25.7%), score 2 (n=332, 16.2%), and score ≥3 (n=186, 9.1%),Fig 1. As shown in Table 1, The mean age of the overall population increased progressively across metabolic risk burden categories, from 39.93±12.30 years in participants with no abnormalities to 49.13±13.57 years in those with ≥3 abnormalities (P<0.001).The proportion of males also rose markedly with increasing metabolic burden,from 31.5% in the lowest group to 86.0% in the highest group (P<0.001).Similarly, indices of adiposity showed a stepwise increase,with BMI rising from 22.05±2.52 kg/m² to 27.70±3.12 kg/m² and waist circumference from 73.60±7.50 cm to 93.35±7.31 cm across categories (both P<0.001).Cardiometabolic parameters also worsened significantly with higher metabolic burden,including increases in systolic blood pressure (106.33±10.77 to 134.14±17.37 mmHg),diastolic blood pressure (66.51±7.60 to 82.55±11.53 mmHg), triglycerides (0.97±0.32 to 3.88±4.57 mmol/L),and fasting blood glucose (4.72±0.40 to 6.10±1.93 mmol/L), along with a progressive decrease in HDL-C levels (1.66±0.32 to 1.26±0.30 mmol/L) (all P<0.001).These findings indicate a clear clustering and progressive worsening of metabolic abnormalities with increasing metabolic risk burden.

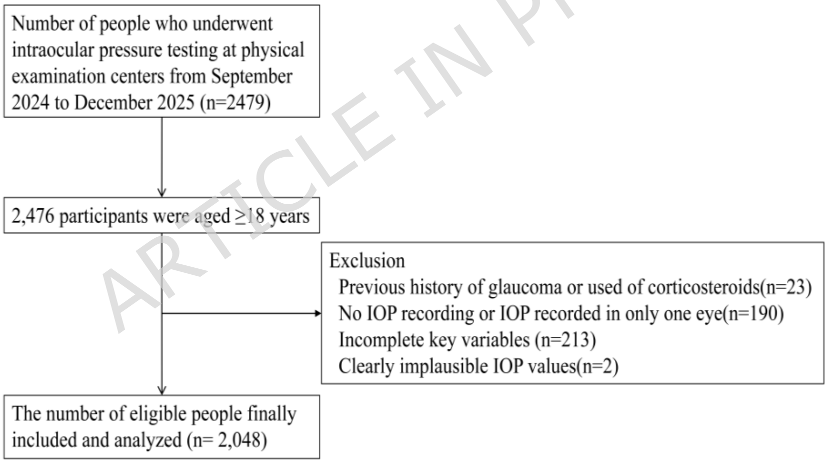


**Figure 1.** selection of study subjects

**Table 1.** Baseline characteristics of participants according to metabolic risk burden categories

| | **0 metabolic abnormalities N=1003(49.0%)** | **1 metabolic abnormality N=527(25.7%)** | **2 metabolic abnormalities N=332(16.2%)** | **⩾3 metabolic abnormalities N=186(9.1%)** | ***P* Value** |
|---|---|---|---|---|---|
| Age(years) | 39.93±12.30 | 46.19±13.44 | 47.98±13.02 | 49.13±13.57 | <0.001 |
| Sex(male,%) | 316 (31.5%) | 331 (62.8%) | 268 (80.7%) | 160 (86.0%) | <0.001 |
| Waist circumference(cm) | 73.60±7.50 | 82.28±8.81 | 88.72±8.63 | 93.35±7.31 | <0.001 |
| BMI(kg/m² ) | 22.05±2.52 | 24.38±3.05 | 26.29±3.23 | 27.70±3.12 | <0.001 |
| Systolic blood pressure(mmHg) | 106.33±10.77 | 120.66±16.36 | 125.12±16.47 | 134.14±17.37 | <0.001 |
| Diastolic blood pressure(mmHg) | 66.51±7.60 | 74.01±10.51 | 77.54±10.72 | 82.55±11.53 | <0.001 |
| Triglycerides(mmol/L) | 0.97±0.32 | 1.71±1.04 | 2.56±2.47 | 3.88±4.57 | <0.001 |

| | | | | | |
|---|---|---|---|---|---|
| HDL-C(mmol/L) | 1.66±0.32 | 1.48±0.32 | 1.35±0.29 | 1.26±0.30 | <0.001 |
| Fasting blood glucose(mmol/L) | 4.72±0.40 | 4.98±0.88 | 5.44±1.43 | 6.10±1.93 | <0.001 |

Values are presented as mean±SD or n (%). P values were obtained using analysis of variance test. Metabolic risk burden categories were defined by the number of abnormal components (0, 1, 2, and ⩾3). BMI, body mass index; SBP, systolic blood pressure; DBP, diastolic blood pressure; TG, triglycerides; HDL-C, high-density lipoprotein cholesterol.

In the unadjusted model (Model 1), mean IOP increased monotonically with higher metabolic burden (group means for scores 0 to≥3: 13.91, 14.70, 14.96, and 15.49 mmHg, respectively; P<0.001). After adjustment for age and sex (Model 2), the increasing trend remained evident (estimated marginal means: 13.84, 14.76, 15.26, and 15.83 mmHg; *P*<0.001). Further adjustment for BMI (Model 3) attenuated the association but did not eliminate it (estimated marginal means: 13.95, 14.72, 14.88, and 15.33 mmHg; *P*<0.001), indicating significant differences in IOP across metabolic risk burden categories (Fig 2 and Table 2).

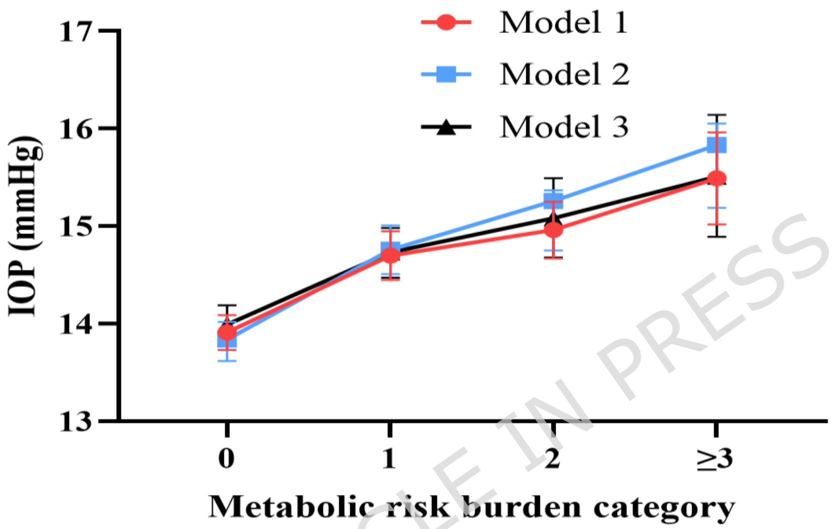


**Figure 2.** illustrates a clear dose-response relationship between metabolic risk burden categories (0, 1, 2, and ⩾3 abnormalities) and IOP. In the crude analysis (Model 1), mean IOP increased progressively with higher metabolic burden. After adjustment for age and sex (Model 2), the upward trend persisted across categories, with consistently higher IOP estimates in the higher-burden groups.

**Table 2.** Mean intraocular pressure across metabolic risk burden categories in crude and adjusted models

| | | **0 abnormalities N=1003** | **1 abnormality N=527** | **2 abnormalities N=332** | **⩾ 3abnormalities N=186** | ***P* Value** |
|---|---|---|---|---|---|---|
| Mean intraocular pressure(mmHg) | Model 1 | 13.91 ± 0.09 | 14.70± 0.13 | 14.96 ± 0.15 | 15.49 ± 0.24 | <0.001 |
| | Model 2 | 13.84± 0.10 | 14.76±0.13 | 15.26±0.20 | 15.83±0.31 | <0.001 |
| | Model 3 | 13.99±0.11 | 14.73±0.13 | 15.08± 0.21 | 15.51 ± 0.32 | <0.001 |

Values are presented as estimated marginal means±SE. Model 1, crude means; Model 2, adjusted for age and sex; Model 3, additionally adjusted for BMI. Post hoc pairwise comparisons were performed with Bonferroni correction.

Multivariable linear regression analyses corroborated these findings (Table 3). After adjustment for age and sex, compared with participants with a score of 0, IOP was higher by 0.93 mmHg, 1.24 mmHg, and 1.80 mmHg in those with scores of 1, 2, and≥3, respectively (all *P*<0.001). With additional adjustment for BMI, the effect estimates were attenuated (β=0.77, 0.93, and 1.37 mmHg, respectively), but remained statistically significant (all *P*<0.001). Notably, Age was inversely associated with IOP in both models (β= −0.02 mmHg per year, P<0.001), whereas sex was not significantly associated with IOP. BMI itself was positively associated with IOP: each 1 kg/$m^2$ increase in BMI corresponded to a 0.09-mmHg higher IOP (95% CI: 0.04-0.13; *P*<0.001).Trend analyses showed that each one-category increase in metabolic risk burden was associated with a 0.61-mmHg and 0.46-mmHg higher IOP in Model 2 and Model 3, respectively, with significant linear trends (both P-trend<0.001), supporting a clear dose-response relationship between metabolic risk burden and IOP.

**Table 3.** Association between metabolic risk burden categories and intraocular pressure: multivariable linear regression

| **Predictor** | **Model 2 β (95% CI)** | ***P* Value** | **Model 3 β (95% CI)** | ***P* Value** |
|---|---|---|---|---|
| Burden1 vs 0 | 0.93(0.61,1.26) | <0.001 | 0.77(0.43,1.10) | <0.001 |
| Burden2 vs 0 | 1.24(0.85,1.63) | <0.001 | 0.93(0.51,1.35) | <0.001 |
| Burden⩾3 vs 0 | 1.80(1.32,2.28) | <0.001 | 1.37(0.84,1.90) | <0.001 |
| Age(per 1-year increase) | -0.02(-0.03,-0.01) | <0.001 | -0.02(-0.03,-0.01) | <0.001 |
| Male(vs female) | 0.00(-0.27,0.28) | 0.988 | -0.12(-0.40,0.16) | 0.400 |
| BMI(per 1kg/$m^2$ increase) | | | 0.09(0.04,0.13) | <0.001 |
| *P* for trend (per 1-category increase) | 0.61 (0.47, 0.75) | <0.001 | 0.46 (0.30, 0.62) | <0.001 |

Data are presented as β (95% CI). Metabolic risk burden categories were entered as indicator variables with 0 abnormalities as the reference. Model 2 was adjusted for age and sex; Model 3 was additionally adjusted for BMI. CI, confidence interval; BMI, body mass index.

In the sensitivity analysis, high IOP was defined as an IOP value at or above the 90th percentile (P90) of the study population. After adjustment for age and sex (Model 2), participants with 1, 2, and ⩾3 metabolic abnormalities had progressively higher odds of high IOP compared with those without abnormalities, with odds ratios (ORs) of 1.86 (95% CI: 1.25-2.75), 2.36 (95% CI: 1.49-3.73), and 4.15 (95% CI: 2.52-6.84), respectively(Table 4).Further adjustment for BMI (Model 3) did not materially alter the associations, although the effect estimates were modestly attenuated ORs: 1.63 (95% CI: 1.08-2.45), 1.86 (95% CI: 1.13-3.08), and 3.05 (95% CI: 1.73-5.36). Age was inversely associated with high IOP in both models, whereas sex was not significantly associated. In Model 3, BMI showed an independent positive association with high IOP (OR=1.06, 95% CI: 1.01-1.11).To further explore potential effect modification, an interaction term between metabolic risk burden and sex was included in the multivariable model; however, no significant interaction was observed (*P* for interaction=0.208).

**Table 4.** Sensitivity analysis: association between metabolic risk burden and high intraocular pressure (IOP) defined as ⩾P 90 of IOP

| **Predictor** | **Model 2 OR (95% CI)** | ***P* Value** | **Model 3 OR (95% CI)** | ***P* value** |
|---|---|---|---|---|
| Metabolic risk burden | | | | |
| 0 | | | | |
| 1 | 1.855(1.250,2.751) | 0.002 | 1.628(1.080,2.453) | 0.020 |
| 2 | 2.355(1.488,3.728) | <0.001 | 1.864(1.128,3.080) | 0.015 |
| ⩾3 | 4.151(2.519,6.839) | <0.001 | 3.045(1.729,5.361) | <0.001 |
| Age (per 1-year increase) | 0.982(0.970,0.994) | 0.003 | 0.983(0.972,0.995) | 0.007 |
| Sex (female vs male) | 1.200(0.856,1.682) | 0.291 | 1.292(0.917,1.821) | 0.143 |
| BMI(per 1kg/$m^2$ increase) | | | 1.059(1.008,1.113) | 0.023 |

High IOP was defined as IOP⩾the 90th percentile (P90). P90=18.25mmHg.Model 2 was adjusted for age and sex; Model 3 was additionally adjusted for BMI. CI, confidence interval; BMI, body mass index.

To further delineate the independent contributions of individual metabolic components to IOP, we simultaneously entered the five components-central obesity, elevated triglycerides, low HDL-C, elevated blood pressure, and elevated fasting glucose-as binary variables into a multivariable linear regression model, adjusting for age and sex (Table 5). After mutual adjustment, elevated blood pressure showed the strongest independent association with IOP (B=1.066±0.161 mmHg; β=0.154; *P*<0.001), followed by elevated fasting glucose (B=1.124±0.249 mmHg; β=0.102; *P*<0.001). Elevated triglycerides (B=0.427±0.155 mmHg; β=0.066; *P*=0.006) and central obesity (B=0.331±0.165 mmHg; β=0.047; P=0.044) were also significantly associated with higher IOP. In contrast, low HDL-C was not significantly associated with IOP (B=0.061±0.290 mmHg; *P*=0.833). Within the same model, age was inversely associated with IOP (per 1-year increase: B=−0.029±0.005 mmHg; β=−0.131; *P*<0.001), whereas sex was not a significant predictor (*P*=0.500). VIFs ranged from 1.08 to 1.23, indicating no evidence of problematic multicollinearity.

**Table 5.** Independent associations of five metabolic components with IOP in multivariable linear

regression

| Predictor | B (mmHg) ± SE | Standardized β | *P* value | VIF | Part r | Part $r^2$ |
|---|---|---|---|---|---|---|
| Age (per 1-year increase) | -0.029±0.005 | -0.131 | <0.001 | 1.155 | -0.122 | 0.0149 |
| Sex (female vs male) | -0.094 ± 0.140 | -0.016 | 0.500 | 1.220 | -0.015 | 0.0002 |
| waist circumference(⩾90 cm for men and ⩾85 cm for women) | 0.331 ± 0.165 | 0.047 | 0.044 | 1.178 | 0.043 | 0.0018 |
| Hypertriglyceridemia(TG ⩾1.7 mmol/L) | 0.427 ± 0.155 | 0.066 | 0.006 | 1.231 | 0.059 | 0.0035 |
| HDL-C(HDL-C<1.04 mmol/L) | 0.061 ± 0.290 | 0.005 | 0.833 | 1.076 | 0.005 | 0.0000 |
| SBP ⩾130 and/or DBP ⩾85 mmHg or treated | 1.066 ± 0.161 | 0.154 | <0.001 | 1.162 | 0.143 | 0.0204 |
| FPG ⩾6.1 mmol/L or treated | 1.124 ± 0.249 | 0.102 | <0.001 | 1.098 | 0.097 | 0.0094 |

Multivariable linear regression with IOP as the dependent variable. All five metabolic components were entered simultaneously and adjusted for age and sex. B represents adjusted mean difference in IOP (mmHg) for presence vs absence of each component. Part $r^2$= (Part r)$^2$; VIF indicates multicollinearity.

## Discussion

With increases in life expectancy and shifts in dietary patterns, the health consequences of metabolic dysregulation have become increasingly apparent.Beyond cardiovascular and cerebrovascular disease, the link between MetS and IOP regulation has attracted growing attention, with substantial economic implications for both individuals and society[16,17]. Using real-world data from routine health examinations, we developed a metabolic risk burden score based on five commonly available components-waist circumference,blood pressure,fasting glucose, triglycerides, and HDL-C. We observed that higher metabolic burden was associated with higher IOP in a clear dose-response manner. Prior studies have often operationalized MetS as a binary exposure ("present" vs "absent") and have consistently reported significantly higher IOP among individuals with MetS[5,12,13]. Building on this evidence, our analysis provides a more granular characterization: even after adjustment for age, sex, and further for BMI, IOP increased stepwise as the number of abnormal metabolic components accumulated. The trend analysis quantified this relationship, indicating that each one-point increase in metabolic burden corresponded to an average 0.46-mmHg higher IOP.These findings support the use of routinely collected metabolic parameters in health check-up settings to stratify the risk of elevated IOP among ostensibly healthy individuals. Notably, the association remained evident after adjustment for age and sex, and persisted-albeit attenuated-after additional control for BMI, suggesting that metabolic clustering may be linked to higher IOP through pathways that are not fully explained by general adiposity alone. Most previous studies have compared MetS-positive versus MetS-negative individuals; while such work has established associations with elevated IOP and increased glaucoma risk, it is less informative regarding whether a graded risk is already present before individuals meet formal diagnostic thresholds for MetS[12,13,18].

By adopting an accumulation-based metabolic scoring approach, our study delineated the full trajectory of IOP changes as metabolic abnormalities increased from a single component to a clustering of three or more components. Importantly, the data suggest that even in individuals who do not meet formal diagnostic criteria for MetS, the aggregation of metabolic abnormalities itself may confer an increased risk of higher IOP.Interestingly, an inverse association between age and metabolic risk burden was observed in this study, with younger individuals more likely to present with a higher number of metabolic abnormalities.This finding differs from the commonly reported increase in metabolic disorders with advancing age in the general population[5,15].Several explanations may account for this observation. First, selection bias may be present in health examination populations, as older individuals who undergo routine check-ups may be more health-conscious and better managed for metabolic conditions. Second, survivor bias may play a role, as individuals with more severe metabolic disorders may be underrepresented in older age groups. Third, recent epidemiological trends suggest an increasing prevalence of metabolic abnormalities among younger adults, possibly related to lifestyle factors such as diet, physical inactivity, and occupational stress[19-20]. In addition, health examination cohorts may not fully represent the general population due to differences in health-seeking behavior and baseline characteristics.Several explanations have been proposed, Age-related changes in aqueous humor dynamics and potential selection bias in health examination populations[21,22]. This observation was further supported by a sensitivity analysis defining high IOP as values at or above the P90 of the study population, in which cumulative metabolic burden demonstrated a clear dose-response association with high IOP (≥P90), with the highest odds ratio reaching 4.15. This association remained statistically significant after adjustment for age and sex, and persisted after further adjustment for BMI, although the effect estimates were slightly attenuated.Given that central obesity is already included in the metabolic risk burden score, additional adjustment for BMI may introduce a degree of overadjustment; therefore, these results should be interpreted with caution. Nevertheless, the consistency of the findings across models suggests that the observed association is robust and not solely explained by general adiposity.In addition, no significant interaction by sex was observed, suggesting that the association between metabolic burden and IOP was generally consistent across men and women. This finding reduces concerns regarding sex-related effect modification and supports the robustness of the observed association.It should be emphasized that the endpoint of this study was IOP rather than incident glaucoma. Accordingly, the primary contribution of our findings is to provide a clinically relevant signal for risk awareness and stratification, rather than diagnostic evidence or direct causal inference regarding glaucoma outcomes. When evaluating the independent associations of individual metabolic components, elevated blood pressure and elevated fasting glucose emerged as the two factors most strongly related to IOP, consistent with findings reported by Yasukawa and Parajuli and colleagues[23,24]. From a mechanistic perspective, metabolic dysregulation may influence IOP regulation through multiple converging pathways. Hypertension, diabetes, and lipid metabolic disturbances have each been implicated in IOP elevation[5,21]. Potts et al. reported that glucagon-like peptide-1 receptor agonists exert beneficial effects-such as attenuating inflammation, promoting weight loss, and reducing cardiovascular mortality-which highlights the interplay between metabolic regulation,systemic vascular and inflammatory pathways[25]. In addition, insulin resistance and sympathetic nervous system activation have been proposed to increase aqueous humor production or raise outflow resistance, thereby contributing to higher IOP[25,26]. Lee et al. further observed that dyslipidaemia in adolescents was significantly associated with the development and progression of open-angle glaucoma[27]. In our mutually adjusted models, elevated triglycerides and central obesity were also independently associated with higher IOP, whereas HDL-C showed no independent association. By contrast, a cross-sectional study in a southern Chinese population reported that lower HDL-C was a risk factor for elevated IOP; differences in population structure and confounder control may partly account for the inconsistent findings[28]. Current evidence generally supports the notion that dyslipidaemia and hypertension contribute to endothelial dysfunction, chronic vascular inflammation, and oxidative stress, which may disrupt the microenvironment and function of the ciliary body and trabecular meshwork[29-31]. Moreover,central obesity may increase intra-abdominal pressure and promote adipokine imbalance, potentially elevating episcleral venous pressure and thereby increasing IOP[12,32,33]. After additional adjustment for BMI, the association between metabolic burden and IOP was attenuated but remained statistically significant, suggesting that obesity is an important mediator but does not fully explain the observed relationship. This pattern aligns with evidence from a longitudinal cohort study in Korea[34]. A substantial body of literature has also demonstrated a positive association between obesity and IOP[30,35]. One meta-analysis reported that individuals with higher body weight tend to have higher IOP, and that elevated IOP is associated with an increased risk of glaucoma, underscoring the clinical relevance of obesity and metabolic dysregulation as potentially independent contributors to IOP elevation.Collectively, these findings suggest that in clinical practice, physicians should look beyond the binary question of whether a patient meets diagnostic criteria for MetS and also consider the number of abnormal metabolic components, as cumulative metabolic burden may provide a more sensitive signal for early identification of individuals at risk of elevated IOP.

From a public health and health management perspective, waist circumference, blood pressure, glucose, and lipid profiles are already part of routine health examinations. Our findings indicate that clinicians and health managers do not need complex calculations to apply these results in practice. Simply counting the number of abnormal components using routinely available measures allows rapid identification of subgroups with a substantially higher likelihood of elevated IOP. For individuals with ≥2 abnormal components-particularly those with ≥3,even if a single IOP reading falls within the upper end of the normal range, the health examination report should explicitly flag this as a higher-risk profile. In such cases, periodic IOP monitoring should be recommended, and referral to ophthalmology for optic nerve evaluation and visual field testing should be considered when appropriate, to facilitate earlier detection of glaucoma and timely intervention.In addition,by adjusting for BMI,we preliminarily disentangled the relative contributions of metabolic clustering versus general adiposity to IOP, which may provide a basis for future mechanistic investigations.In this cross-sectional study, metabolic risk burden was independently associated with higher IOP in a dose-response manner. These findings suggest a potential relationship between metabolic abnormalities and IOP, but further prospective studies are needed to establish causality and clinical implications.

Several limitations warrant consideration. First, the cross-sectional design precludes causal inference regarding the relationship between metabolic abnormalities and IOP. Residual confounding is also possible, as certain variables were not available in our dataset, including family history, dietary patterns, and other lifestyle factors; prospective studies are needed to confirm these associations.Second, participants were derived from a health examination population, which may introduce selection bias and may not fully represent the general population. In addition, although the majority of participants were from the general population, the single-center design and specific study setting may limit the generalizability of the findings to other populations.Third, IOP was measured using non-contact tonometry within the health examination setting, and measurement variability related to participant status or testing conditions may have introduced random error. Moreover, key ocular parameters influencing IOP assessment, particularly central corneal thickness and axial length, were not available, which may have resulted in residual confounding.Finally, we did not perform heterogeneity analyses across specific combinations of metabolic abnormalities.Future work should explore whether particular components or clustering patterns are more strongly associated with IOP elevation, which could further refine risk stratification in clinical practice.

**Abbreviations**

BMI:body mass index
DBP:diastolic blood pressure
SBP:systolic blood pressure
TG:triglycerides
HDL-C:high-density lipoprotein cholesterol
FPG:fasting plasma glucose
IOP:intraocular pressure
MetS:metabolic syndrome
SD:standard deviation
SE:standard error
CI:confidence interval
EMM:estimated marginal means
OR:odds ratio
P90:90th percentile
P-trend:P value for trend
VIF:variance inflation factor



**Declarations**

**Ethics approval and consent to participate**

This cross-sectional study was approved by the Ethics Committee of the Army Medical Center(Approval No.2026-043). As this study was cross-sectional in nature, used de-identified health examination data, and involved no additional interventions or contact with participants, the Ethics Committee waived the requirement for written informed consent to participate. Our research adhered to the tenets of the Declaration of Helsinki.

**Consent for publication**

Not applicable.

**Availability of data and materials**

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request

**Competing interests**

The authors have declared that no competing interests exist.

**Funding**

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

**Author contributions**

QX initiated the study design; HL had participated in manuscript drafting and editing. CMJ and XTH collected the data; XG and JW have made substantial contributions to the interpretation and statistical analysis of the data; LH,YL and XX wrote the draft. All authors reviewed the study protocol and approved the final manuscript.

**Acknowledgements**

The authors wish to extend their gratitude to the nursing staff of the Department of Health Management at Daping Hospital, Army Medical University, for their unwavering support throughout the data collection period.

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